PASS Data Community Summit

Host redgate

Change Tracking in SQL Server 2025: Exploring Change Event Streaming vs CDC

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- 25 years as a DBA
- Data Platform Women in Tech (WIT) Virtual UG, Co-leader
- WITspiration, Co-founder
- Redgate Community Ambassador
- Microsoft MVP, Data Platform
- In my spare time, I can usually be found doing something musical or something geeky with my husband, Andy, and our dog, Sebastian.

Here's the Scenario:

- Breaking up a legacy monolith system into smaller parts
  - Microservices
  - Many databases
  - Teams choosing their database RDBMS & set ups
- Creation of analytical system
  - (Near) Real-time analytics
- No cross-database joins
- Need to share data across the systems
- Need before and after data changes

Options We're Ruling Out

Option 1: Add columns to the table

Pros:

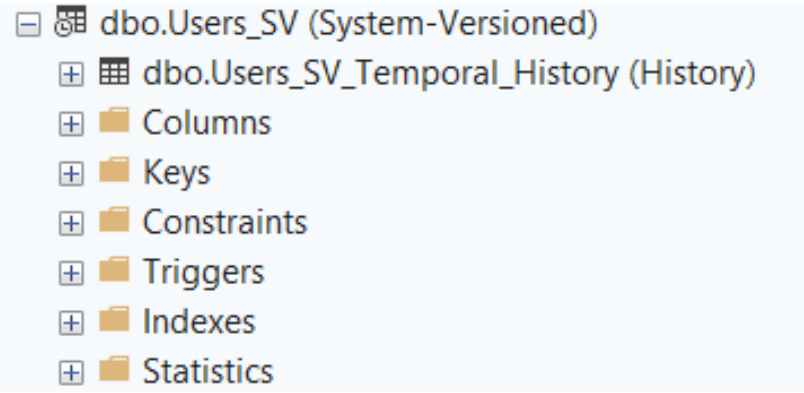
- We can use these columns directly

Cons:

- Querying large tables can take time
- Not guaranteed to always be populated properly
- Only has the updated data

```
CREATE TABLE dbo.Users(
    Id INT IDENTITY(1,1) NOT NULL,
    DisplayName NVARCHAR(40) NOT NULL,
    CreationDate DATETIME NOT NULL,
    LastUpdateDate DATETIME NOT NULL,
    CONSTRAINT PK_Users_Id PRIMARY KEY CLUSTERED (Id),
    INDEX IDX_Users_DisplayName (DisplayName)
    )
```

Option 2: Temporal Tables\System Versioned Tables



Pros:

- Maintained by SQL Server
- Stores both previous and current versions of rows

Cons:

- Querying large tables can take time
- Potential overhead if not set up properly with additional indexing and retention periods.
- See documentation for limitations

Option 3: Change Tracking

Pros:

- Only identifies that row that has changed
- Managed by SQL Server

Cons:

- Does not capture the original or updated data
- Requires additional set up and overhead.

What are we left with?

Change Data Capture (CDC)

- Introduced SQL Server 2008
- “Pull”

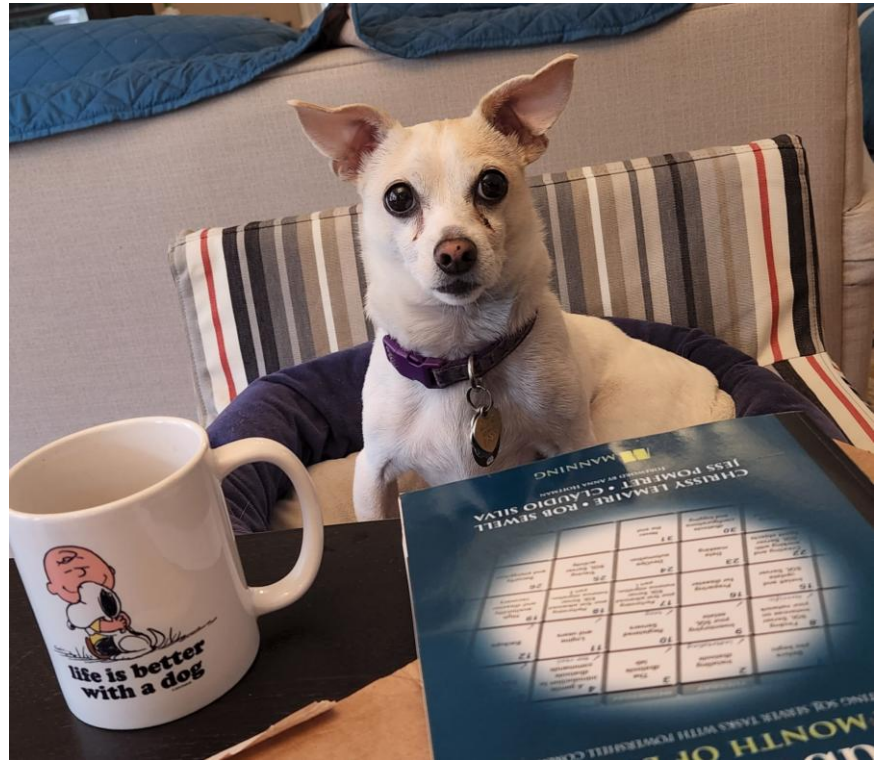
Change Event Streaming (CES)

- Introduced SQL Server 2025
- Still in Preview
- “Push”

Agenda

- How CDC & CES Work
- Setup
- Demos
- “Cautionary Tales”
- When To Use Each

How They Work



Overview of Capture Process for CDC & CES

- Specify tables to track
- Transaction Logs are scanned for changes which are sent to the appropriate system
  - CDC – cdc schema tables inside the database
  - CES – Azure Event Hubs
    - Managed, high-throughput message service
- Logs receive acknowledgement of changes
- After retention period, change data is removed

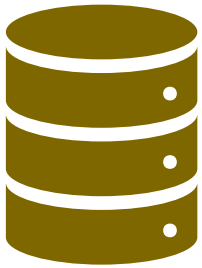
How this works:

```
UPDATE Customer
SET State = 'MA'
WHERE FirstName = 'Sebastian'
```

Fake\Mock Transaction Log

Log Seq Number (LSN)	Table	Record Info	Chg	Clear?
0000000A	Customer	Sebastian-AZ	Old	✓
0000000A	Customer	Sebastian-MA	New	✓

cdc schema tables



CDC

Acknowledgement

OR

Changes Sent

CES

Azure Event Hubs



Transaction Logs – Word of Warning

- Can't clear logs if changes can't be written or messages haven't been acknowledged.
- Monitor Log Growth to make sure that these are not blockers:

```
SELECT db.name,
    db.is_cdc_enabled,
    db.is_change_feed_enabled,
    db.log_reuse_wait_desc,
    usage.total_log_size_in_bytes/1024 AS TotalSizeInKB,
    usage.used_log_space_in_bytes/1024 AS TotalSpaceInKB,
    usage.used_log_space_in_percent,
    usage.log_space_in_bytes_since_last_backup/1024 AS SpaceUsedSinceLastBackupInKB
FROM sys.databases AS db
JOIN sys.dm_db_log_space_usage AS usage ON usage.database_id = db.database_id
```

REPLICATION

Compatibility Matrix: Platform Availability

	SQL Server 2025	Azure SQL Database	Azure SQL MI
CDC	**Yes**	**Yes**[3]	**Yes**
CES	Preview	Preview	Preview[2]
Fabric Mirroring	**Yes (via Azure Arc)**	**Yes**	**Yes**[2]
Transactional replication	**Yes**	Subscriber only[1]	**Yes**

1 Azure SQL Database can only be a transactional replication subscriber, not a publisher or distributor.
2 MI requires the SQL Server 2025 or Always-up-to-date update policy; on the 2022 policy, mirroring uses CDC.
3 CDC runs on any vCore tier for Azure SQL Database; on the DTU model it needs S3 or higher (Basic and S0–S2 are not supported).

Compatibility Matrix: Which Features Can Run Together

	CDC	CES	Fabric Mirroring	Replication
CDC	—	No	No[2]	Yes[1]
CES	No	—	No[3]	No
Fabric Mirroring	No[2]	No[3]	—	No
Replication	Yes[1]	No	No	—

1 On SQL Server 2025 and Azure SQL MI. Azure SQL Database can only be a replication subscriber.
2 On SQL Server 2016–2022 and MI with the 2022 update policy, Fabric Mirroring uses CDC.
3 Documented for SQL Server only.

Setup

CDC

Azure Event Hubs

CES



CDC Set up

- Enable CDC in the database
- Enable CDC for the tables
- SQL Server Agent must be running
  - 2 jobs are created per database: Capture & Cleanup
  - Scheduler for Azure SQL DB

Agent Job Activity:

Name	Enabled	Status	Last Run Outcome	Last Run
cdc.StackOverflow_CDC_capture	yes	Executing...	Unknown	never
cdc.StackOverflow_CDC_cleanup	yes	Idle	Succeeded	3/4/2026 3:07:43 AM

CDC SQL Agent Jobs – Capture

- Settings for job procs
  - MaxTrans = # of transactions in each scan cycle (Default 10000)
  - Maxscans = # of scan cycles to execute to get all the rows (Default 10)
  - Continuous = Is this running continuously? (Default 1)
  - PollingInterval = # of seconds between scans (Default 5)
- Must be running for CDC to be working
- Starts when SQL Agent Job starts by default
  - Consider adding an additional job schedule to start if it is stopped
  - Add check to monitoring\alerting system to see if it's not running

CDC SQL Agent Jobs – Cleanup

- Settings for job procs
  - Retention Period = How long to keep records (Default 4320 minutes\3 days)
  - Threshold = Batch size (Default 5000)

- Deletes in batches:

```
delete top( @p1 )  from [cdc].[dbo_Users_CT]
where __$start_lsn < @p2
```

  - Performance Consequence if not run in a while

- SQL Server – defaults to once a day
- Azure SQL DB - Runs every hour while database is running

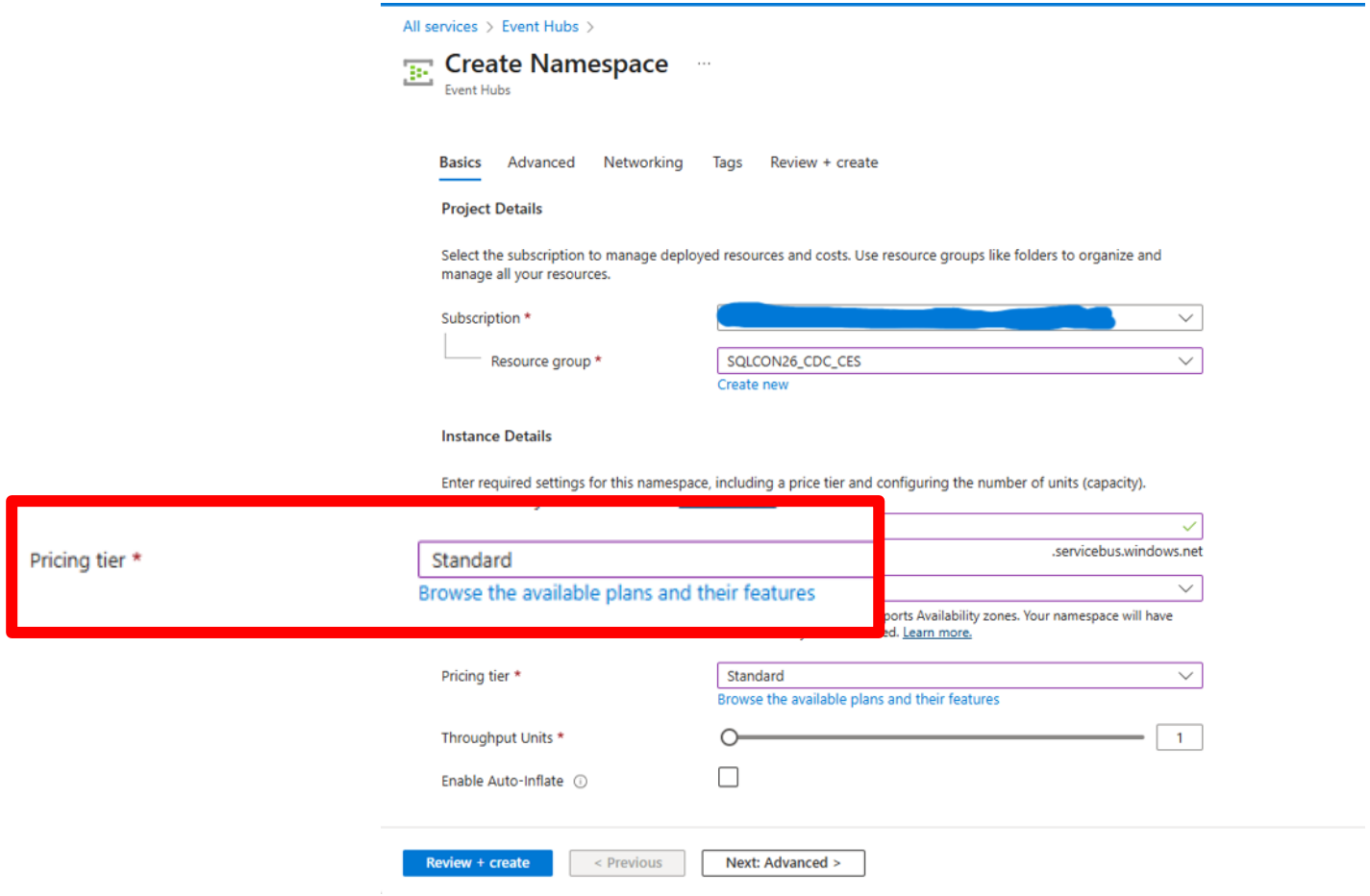
CES Set up - Azure Event Hubs

- Must be set up before CES is enabled
- Choose protocol for the messages
  - ~~AMQP~~ ***This has been deprecated as of August 2026***
  - Kafka
- Security currently differs depending on your SQL engine
  - EntraID (Azure SQL DB & SQL Server with Azure Arc/Azure VM)
  - Security Access Key (SQL Server)
- Can have multiple partitions for messages in Event Hub

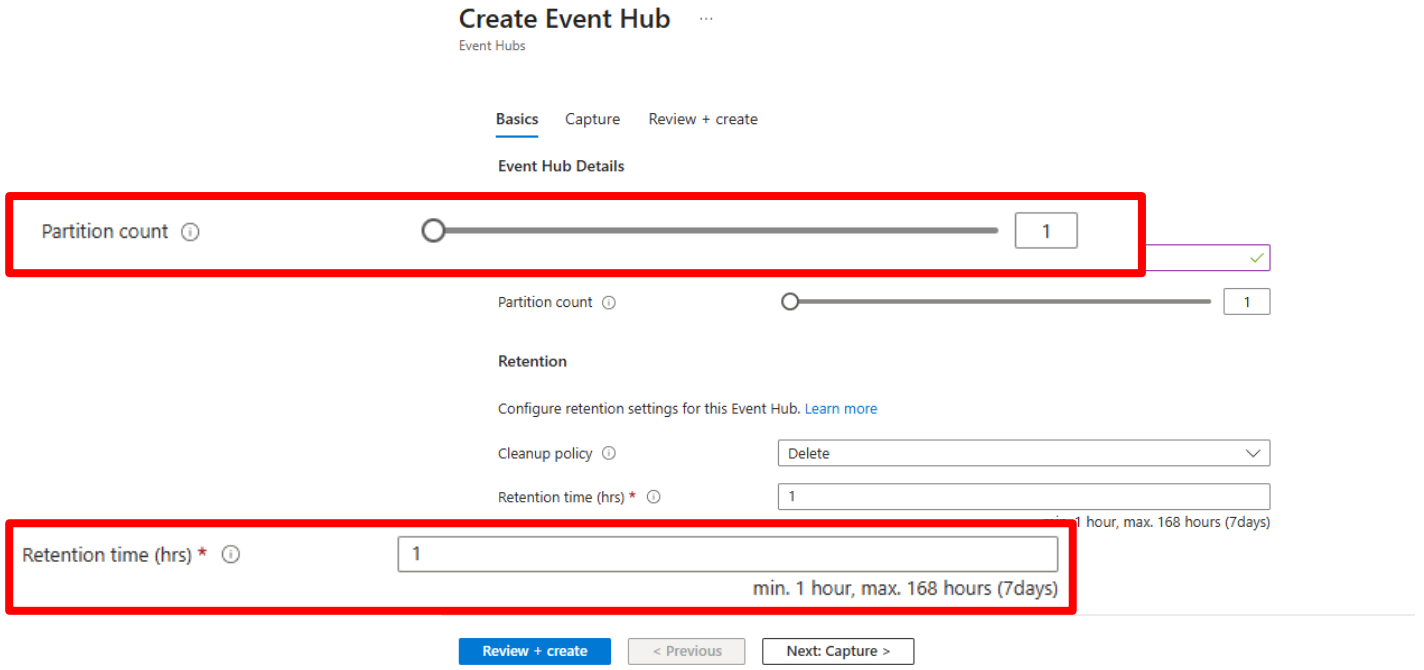
1. Create the Namespace

Pricing Tiers:

- Basic
- Standard
- Premium
- Dedicated



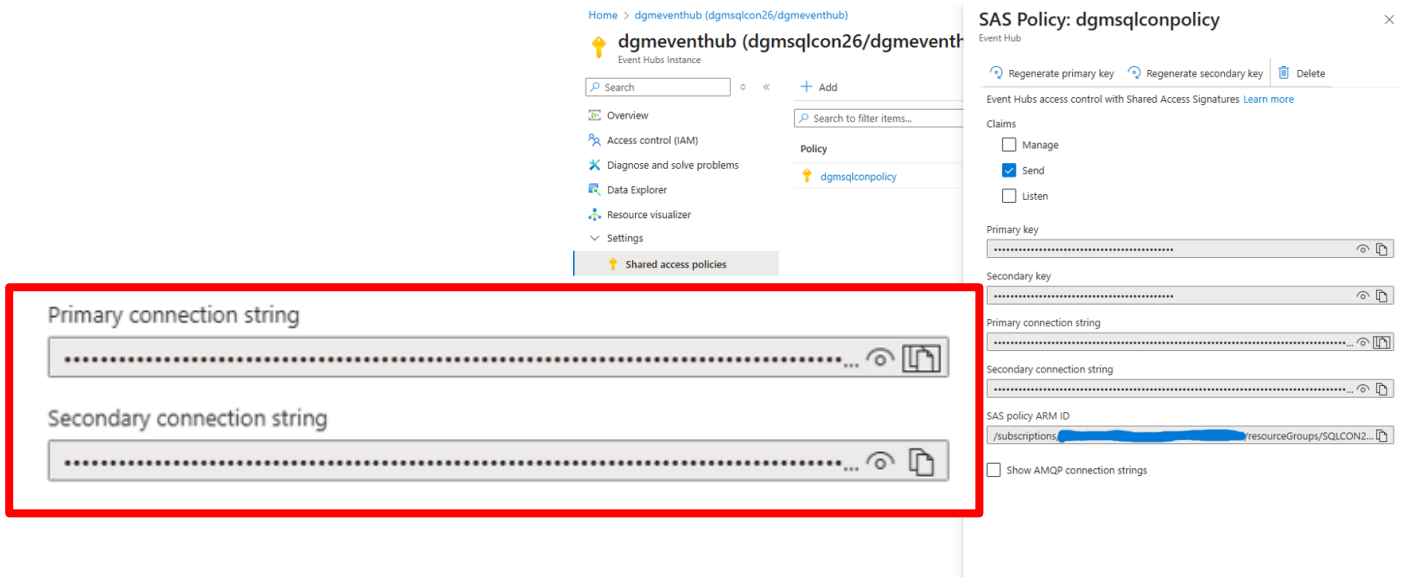
2. Create the Event Hub



3. Create the Shared Access Policy – Kafka Protocol

Skip if using EntraID

(Azure SQL DB or Azure Arc or Azure VM)

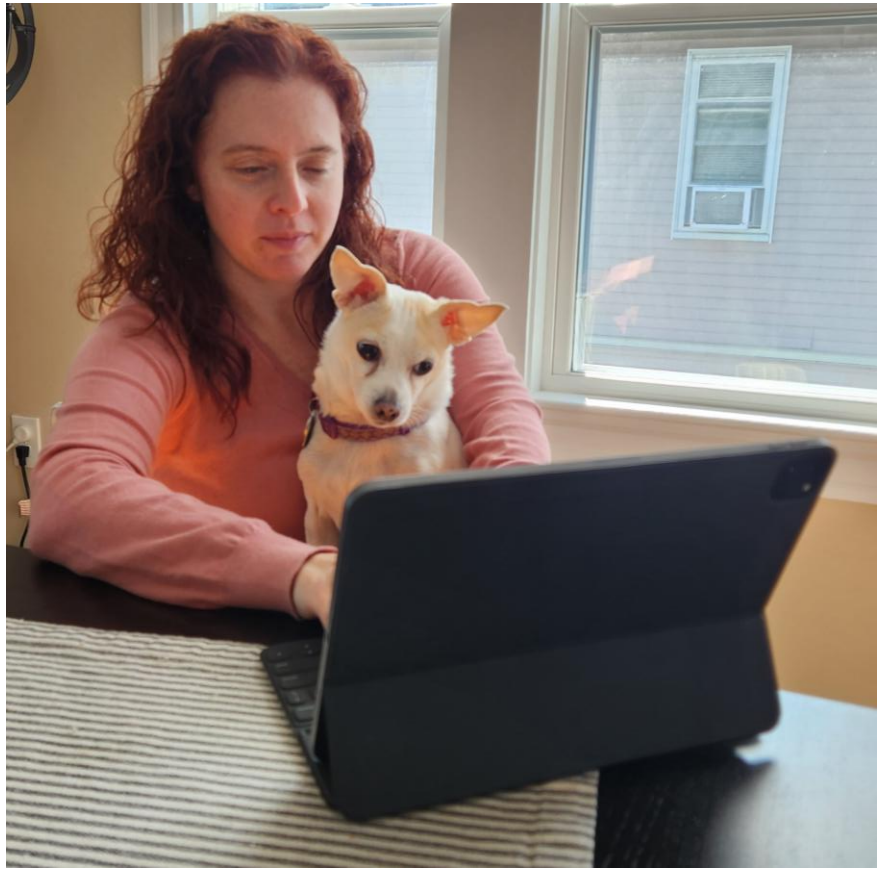


CES Set up - Database

- Enable Preview Features as this is technically still in Preview
  - Not needed for Azure SQL DB
- Enable Event Stream for the database
- Create an Event Stream Group for sending the messages
- Add the table to the stream group
  - Tables can only be in one stream group

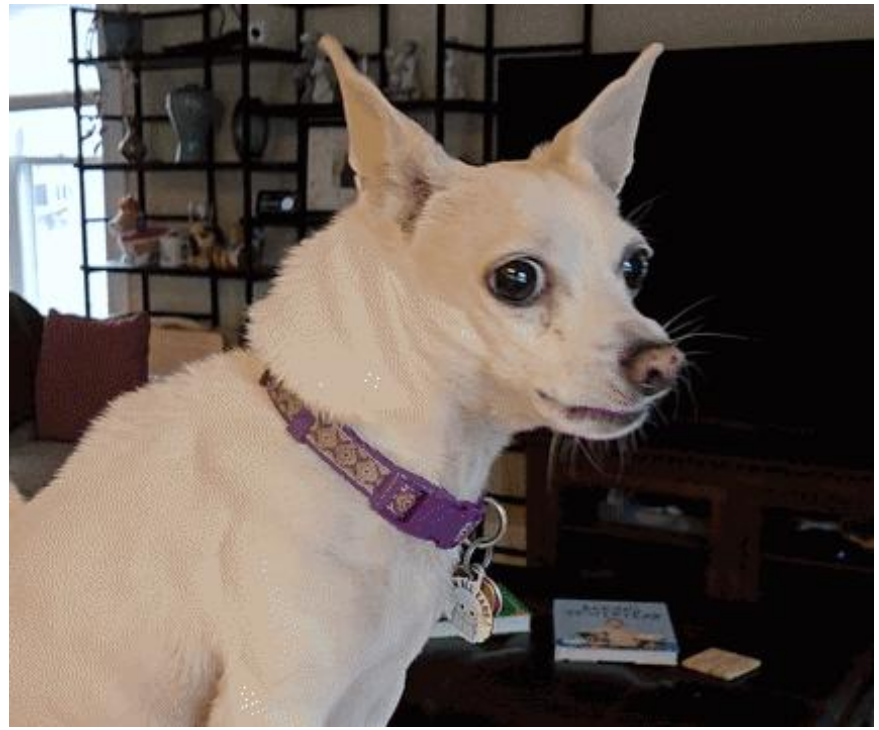
Demos

- Set up database
- Make changes



Cautionary Tales

aka Problems to Look For



High Availability – Availability Groups

- Neither are AG aware
- Consider making modifications to the CDC jobs:
  - Add a job step to make sure that the database is the primary replica.
    - If it is, continue as normal. Otherwise, end job.
  - Add the jobs to the replicas in advance so they are ready to kick off if/when a failover occurs.
- If you are using read replicas to get LSNs for processing, the LSNs could be out of sync if the AGs are out of sync.

Disaster Recovery

- Restores
  - CDC requires an additional command for cdc to be retained: WITH KEEP_CDC
  - CES restores do not keep CES settings
- Disabling CDC erases everything!

Processing Messages (CES)

- Ordering of events isn't necessarily maintained
  - May need additional help to identify the order of the updates
  - Within a partition, the logicalid value may be able to help
- Duplicate Messages are possible
  - If an acknowledgment isn't received for some reason, SQL Server will keep retrying the send
  - Consumer should be aware and know how to handle these cases
- Learning Curve
  - Azure, JSON, .NET, Kafka, Stream Analytics Jobs, etc.

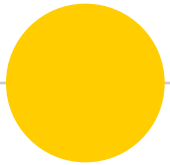
Data Initialization

- CDC and CES are about capturing and streaming changes
- May require full resets at some point for some reason
  - New Setup
  - “Breaking” Schema Changes
  - Disaster Recovery
- Have a plan for reseeding the data for the target systems

Money! Money! Money!

- CES & Azure Event Hubs
  - Pricing Guide: Azure Event Hubs Pricing
  - Make sure you have the right tier set up
  - Know what happens when you hit limits
  - Emulator (unsupported) for development purposes:
    - MS Learn – Azure Event Hubs Emulator Overview

- CDC on Azure SQL DB
  - I/O of writing to and deleting from cdc schema tables
  - Transaction logs



When to use each

When to use CDC

- Changes do not need to be changed outside the environment
  - Same database
  - Different database, same SQL Server instance
  - Different SQL Server instance, same network
- Batch processing
- No Azure or other cloud service
- Need Replication as well
- SQL Server 2022 or earlier (Can still use with SQL 2025)

When to use CES

- Distributed systems
  - Azure DBs only
  - Hybrid Cloud & On Prem
  - Different RDBMS

- Real-time/near real-time processing
  - “Streaming data”

Resources



Transaction Log Resources

- MS Learn – SQL Server Transaction Log Architecture and Management Guide
- Paul Randal Blog Series:
  - https://www.sqlskills.com/blogs/paul/category/transaction-log/
  - https://www.sqlskills.com/blogs/paul/the-sql-server-transaction-log-part-1-logging-basics/

CDC Resources

- MS Learn – About Change Data Capture SQL Server
- MS Learn – Change Data Capture Overview (Azure SQL DB)
- MS Learn – Enable and Disable Change Data Capture – SQL Server
- MS Learn – Administer and Monitor Change Data Capture
- MS Learn – System Tables for Change Data Capture
- MS Learn – System Stored Procs – Change Data Capture

CES & Azure Event Hubs Resources

- MS Learn – Change Event Streaming Overview
- MS Learn – Change Event Streaming – Configure
- MS Learn – Change Event Streaming – FAQ
- MS Learn – Azure Event Hubs – Create
- MS Learn – Azure Event Hubs Emulator Overview
- Azure Event Hubs Pricing

Fabric Mirroring Resources

- MS Learn – Fabric Mirroring Limitations (SQL Server)
- MS Learn – Fabric Mirroring Limitations (Azure SQL Database)

Thank you

Any Final Questions?



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